

# Software Project Proposal

Project Name: uni lms

|                   |                                  |
|-------------------|----------------------------------|
| Prepared For:     | Emaan Swati (ek139426@gmail.com) |
| Industry Domain:  | Education                        |
| Target Platforms: | Web                              |
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## Executive Summary

The proposed project 'uni lms' is designed to serve as a high-quality education platform. Built for web, it incorporates core features including auth, notifications, ai, upload. The proposed EdTech platform delivers a virtual learning environment for students and teachers. It emphasizes course management, content consumption, and interactive quizzes. The solution is optimized for varying network conditions, allowing students to access videos and text materials with minimum buffer times. The backend is structured to support scaling concurrent users during peak exam periods.

## Cost & Timeline Estimates

| Total Est. Effort | Est. Development Cost | Project Timeline | Team Size |
|-------------------|-----------------------|------------------|-----------|
| 264 hours         | \$26,400.00           | 4-6 weeks        | 2 members |

**Recommended Project Roles:** Frontend Developer, Backend Developer

## Scope of Work & Features

| Feature Name          | Description                                                                           |
|-----------------------|---------------------------------------------------------------------------------------|
| Authentication        | Secure user sign up, login, password recovery, and user profiles.                     |
| Push Notifications    | In-app alerts, mobile device registration, and notification history.                  |
| AI Recommendations    | Intelligent machine-learning matching, recommendations engine, and smart text output. |
| Media Storage         | Cloud file uploading, file organization directories, and media compression.           |
| Advanced Search       | Faceted indexing, filtering, search keywords, and high-speed retrieval.               |
| Dashboard & Analytics | Interactive metrics, graphical data charts, and usage analysis.                       |

## Recommended Technology Stack

| Layer                    | Recommended Technology                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------|
| Frontend                 | React + Vite (Tailwind CSS, Framer Motion)                                              |
| Backend                  | FastAPI (Python), WebSockets for real-time quiz updates and chat rooms.                 |
| Database                 | MongoDB or PostgreSQL with JSON fields for flexible course curricula.                   |
| Hosting & Infrastructure | Cloudflare CDN for video caching, AWS S3 for media storage, Vercel/AWS Amplify.         |
| External Integrations    | Zoom API for live lectures, Vimeo API for video hosting, Zoom SDK for inside-app calls. |

## Technical Risks & Mitigations

| Identified Risk Factor                                 | Proposed Mitigation Strategy                                                                             |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Video streaming buffering on low bandwidth connections | Implement adaptive bitrate streaming (HLS/DASH) and utilize Cloudflare CDN.                              |
| Cheating on quizzes and assignments                    | Randomize question orders, disable copy-paste actions on quiz pages, and add strict submission timeouts. |
| Media storage costs rising quickly with course uploads | Apply automatic video compression and transcoding tools (AWS Elemental MediaConvert) on upload.          |